Lab 3

Controlling the 7-Segment Displays with Digital Logic

1. Introduction

A seven-segment display is an electronic component that can be used to display a 1-digit number from 0-9 and a letter *A* to *F*. In this lab we will design a digital circuit to display digits 0 to 9 only, as shown in Figures 1. The circuit we will design is called a *Binary Coded Decimal* (BCD) decoder, Figures 2. This circuit takes in a 4-bit input representing the binary equivalent of a digit 0 to 9 and lights up the desired segments to display the corresponding digit. You are **not** allowed to use any of the Quartus software library modules unless for the modules that are explicitly allowed in the lab instructions.

Figure 1: 7-segment display output digits

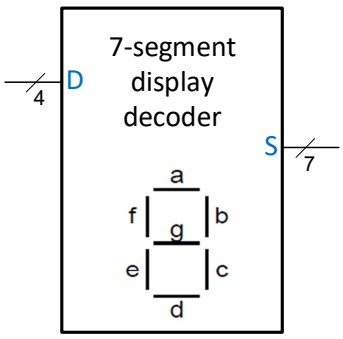

Figure 2: 7-segment Decoder

1.1 Pre-Lab Assignment

The DE1-SoC FPGA circuit turns **on** any segment in the 7-segment display by applying logic 0 to that segment (i.e., active low) . To turn it off, apply logic 1. This is because the 7-segment LED displays have a *common anode*, which means the anode (positive) side of all of these LEDs are electrically connected together to the Vcc (logic 1), and each LED cathode (negative) side is connected to the input pin that drives the LED, as shown in Figure 3.

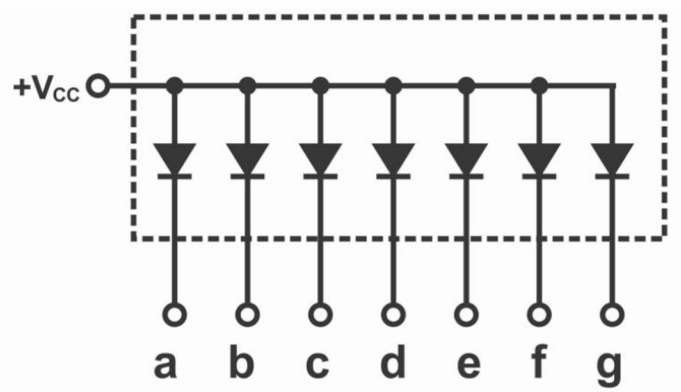

Figure 3: Common anode LEDs configuration

1. Complete the table below to indicate which segments will be turned **ON (0)** vs **OFF (1)** for the digits 0 to 9 shown in Figure 1 above. The entries for digit 0 have been done for you.

Decimal #	BCD Inputs				7-Segment Outputs						
	D_3	D_2	D_1	D_0	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	1	0	0	1	1	1	1
2	0	0	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	1	1	0
4	0	1	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	1	0	0	1	0	0
6	0	1	1	0	0	1	0	0	0	0	0
7	0	1	1	1	0	0	0	1	1	0	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	0	1	0	0

Table 1: Digits to display on the 7-segment display

2. Use seven Karnaugh maps to find the minimized Boolean expressions for each of the seven outputs shown in Table 1 above (outputs ***a*** to ***g***).

Include the answers to pre-lab assignments 1 and 2 above in your report submission. Have their requirements ready to be checked at the beginning of the lab session.

2. Testing the BCD System on the DE1-SoC Board

1. Create a new folder for a new project named **display7** on your computer. Create the new project following the steps outlined in the *Quartus Prime Lite Reference* (in Canvas: Lab Resources). Set the project folder to the **display7** folder and project name to **display7**.
2. On Quartus Schematic, build the BCD digital circuits (four inputs and seven outputs) based on your Boolean expressions from the pre-lab assignment.
3. Run your design on the DE1-SoC board and test it on the first 7-segment display (HEX0).
   a. Use the first four switches, **SW[3]**, **SW[2]**, **SW[1]**, and **SW[0]** (listed on Table 3-6 below) to represent the 4-bit inputs **D3, D2, D1, D0**, respectively.
   b. Use the **HEX0[0]**, **HEX0[1]**, … , **HEX0[6]** pins (listed on Table 3-9 below) to represent the 7-segment outputs **a**, **b**, …, **f** respectively.

   Refer to the *Quartus Prime Lite Reference* (available on Canvas under Lab Resources), for the steps.

Assignment 1
Verify the correct behavior and output on the display by changing the switches settings. Take pictures of your Board's switches and 7-segment displays with at least six different settings for the switches and show the corresponding output on the display. Include the pictures in your report.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allows saving pictures for print purposes.

3. 7-Segment Display with Enable and Negative Inputs

In this part of the lab, you will modify the 7-segment decoder to have two new inputs: ***En*** and ***Ng*** as shown in Figure 4. In the current implementation, when all the four switches of the ***D*** input are OFF, representing number zero, the display shows the digit '0' which is the expected behavior so far. In some applications when the display is not in use, we want it to have all its segments turned OFF. The new ***En*** (enable) input to our BCD digital circuits has been added to help us with this option. When the ***En*** switch input is ON (i.e., binary 1), the 7-Segment display must work as usual by displaying the decimal digit corresponding to 4-bit binary input ***D***. When the ***En*** input is OFF (i.e., binary 0), all displays must be turned OFF and must not show anything regardless of the value on input ***D***.

You will use the other new input ***Ng*** (negative) to display the negative sign. When the switch of this input is ON (i.e., binary 1), the 7-Segment display shows only the negative sign (i.e., only segment ***g*** is ON). When the **Ng** input is OFF (i.e., binary 0), the 7-Segment display must work as usual (i.e., when enabled, they display the decimal digit corresponding to 4-bit binary input ***D***).

1. Create a new folder for a new project named **display7_en** on your computer. Create the new project and set the project folder to the **display7_en** folder and project name to **display7_en**.
2. Copy and paste the schematic of the **display7** module you implemented in the previous assignment. Add to it the *Enable* and *Negative* inputs with the necessary logic gates to function as described above.
3. Compile and upload this circuit to your DE1-SoC board and test it. Use **SW[9]** for the ***En*** input and **SW[4]** for the ***Ng*** input.

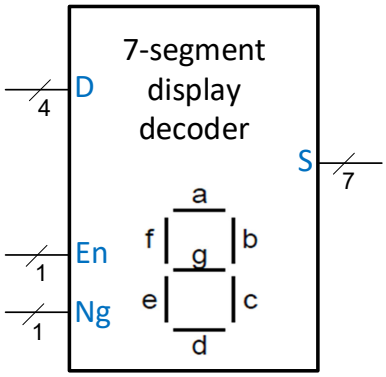

Figure 4: Updated Decoder

4. Pack the *display7_en* schematic as a logic block named **display7_en.bsf** for later use. To do this, go back to your *display7_en* schematic and on the **File** main menu, select **Create/Update → Create Symbol Files for Current File** and save the schematic as **display7_en.bsf** file.

Assignment 2

Take a screenshot of your design and include it in the lab report. Take pictures of your Board's switches (including the enable and negative switches) and the 7-segment displays with at least four different settings of the switches, which must show the effect of the ***En*** and ***Ng*** inputs, and the corresponding outputs on the display. Include the pictures in your report.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allows saving pictures for print purposes.

4. Controlling Multiple 7-Segment Displays with Enable

The DE1-SoC features six 7-Segment displays. In this part of the lab, you will display a digit on two 7-segment displays if they are enabled. The digit will be represented by the first four switches SW[3:0], while SW[4] will be used for the negative sign display. Switches SW[8] and SW[9] will be used to enable the 7-segment displays HEX0 and HEX1 respectively.

1. Create a new folder for a new project named **disp7_control** on your computer. Copy the **display7_en.bdf**, and **display7_en.bsf** files to this new project folder.
2. Create the new project and set the project folder to this **disp7_control** folder and project name to **disp7_control**. Refer to the steps in *Quartus Prime Lite Reference*.
3. Use the first four switches, **SW[3], W[2]**, **SW[1]**, and **SW[0]** as the input representing digits 0 to 9 and switch **SW[4]** for the negative sign for both displays. Switches **SW[9]** and **SW[8]** are used to enable and disable each of the displays **HEX1,** and **HEX0** respectively.
4. Design the system such that the decimal digit or the negative sign are displayed on a display if enabled. If disabled, the display should be turned OFF.
5. Upload your design into the DE1-SoC board and test it on the 7-Segment displays.

Assignment 3

Verify the correct behavior and output on the displays by changing the inputs on the switches for the digits, sign, and enable inputs. Take pictures of your Board's switches and 7-segment displays with at least six different settings for the switches and corresponding output on the displays. Include the pictures in your report.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allows saving pictures for print purposes.

Table 3-6 Pin Assignment of Slide Switches

Signal Name	FPGA Pin No.	Description	I/O Standard
SW[0]	PIN_AB12	Slide Switch[0]	3.3V
SW[1]	PIN_AC12	Slide Switch[1]	3.3V
SW[2]	PIN_AF9	Slide Switch[2]	3.3V
SW[3]	PIN_AF10	Slide Switch[3]	3.3V
SW[4]	PIN_AD11	Slide Switch[4]	3.3V
SW[5]	PIN_AD12	Slide Switch[5]	3.3V
SW[6]	PIN_AE11	Slide Switch[6]	3.3V
SW[7]	PIN_AC9	Slide Switch[7]	3.3V
SW[8]	PIN_AD10	Slide Switch[8]	3.3V
SW[9]	PIN_AE12	Slide Switch[9]	3.3V

Ref: DE1-SoC_User_Manual under
"Reference Material on Canvas"

Table 3-9 Pin Assignment of 7-segment Displays

Signal Name	FPGA Pin No.	Description	I/O Standard
HEX0[0]	*PIN_AE26*	*Seven Segment Digit 0[0]*	*3.3V*
HEX0[1]	*PIN_AE27*	*Seven Segment Digit 0[1]*	*3.3V*
HEX0[2]	*PIN_AE28*	*Seven Segment Digit 0[2]*	*3.3V*
HEX0[3]	*PIN_AG27*	*Seven Segment Digit 0[3]*	*3.3V*
HEX0[4]	*PIN_AF28*	*Seven Segment Digit 0[4]*	*3.3V*
HEX0[5]	*PIN_AG28*	*Seven Segment Digit 0[5]*	*3.3V*
HEX0[6]	*PIN_AH28*	*Seven Segment Digit 0[6]*	*3.3V*
HEX1[0]	*PIN_AJ29*	*Seven Segment Digit 1[0]*	*3.3V*
HEX1[1]	*PIN_AH29*	*Seven Segment Digit 1[1]*	*3.3V*
HEX1[2]	*PIN_AH30*	*Seven Segment Digit 1[2]*	*3.3V*
HEX1[3]	*PIN_AG30*	*Seven Segment Digit 1[3]*	*3.3V*
HEX1[4]	*PIN_AF29*	*Seven Segment Digit 1[4]*	*3.3V*
HEX1[5]	*PIN_AF30*	*Seven Segment Digit 1[5]*	*3.3V*
HEX1[6]	*PIN_AD27*	*Seven Segment Digit 1[6]*	*3.3V*
HEX2[0]	*PIN_AB23*	*Seven Segment Digit 2[0]*	*3.3V*
HEX2[1]	*PIN_AE29*	*Seven Segment Digit 2[1]*	*3.3V*
HEX2[2]	*PIN_AD29*	*Seven Segment Digit 2[2]*	*3.3V*
HEX2[3]	*PIN_AC28*	*Seven Segment Digit 2[3]*	*3.3V*
HEX2[4]	*PIN_AD30*	*Seven Segment Digit 2[4]*	*3.3V*
HEX2[5]	*PIN_AC29*	*Seven Segment Digit 2[5]*	*3.3V*
HEX2[6]	*PIN_AC30*	*Seven Segment Digit 2[6]*	*3.3V*
HEX3[0]	*PIN_AD26*	*Seven Segment Digit 3[0]*	*3.3V*
HEX3[1]	*PIN_AC27*	*Seven Segment Digit 3[1]*	*3.3V*
HEX3[2]	*PIN_AD25*	*Seven Segment Digit 3[2]*	*3.3V*
HEX3[3]	*PIN_AC25*	*Seven Segment Digit 3[3]*	*3.3V*
HEX3[4]	*PIN_AB28*	*Seven Segment Digit 3[4]*	*3.3V*
HEX3[5]	*PIN_AB25*	*Seven Segment Digit 3[5]*	*3.3V*
HEX3[6]	*PIN_AB22*	*Seven Segment Digit 3[6]*	*3.3V*
HEX4[0]	*PIN_AA24*	*Seven Segment Digit 4[0]*	*3.3V*
HEX4[1]	*PIN_Y23*	*Seven Segment Digit 4[1]*	*3.3V*
HEX4[2]	*PIN_Y24*	*Seven Segment Digit 4[2]*	*3.3V*
HEX4[3]	*PIN_W22*	*Seven Segment Digit 4[3]*	*3.3V*
HEX4[4]	*PIN_W24*	*Seven Segment Digit 4[4]*	*3.3V*
HEX4[5]	*PIN_V23*	*Seven Segment Digit 4[5]*	*3.3V*
HEX4[6]	*PIN_W25*	*Seven Segment Digit 4[6]*	*3.3V*
HEX5[0]	*PIN_V25*	*Seven Segment Digit 5[0]*	*3.3V*
HEX5[1]	*PIN_AA28*	*Seven Segment Digit 5[1]*	*3.3V*
HEX5[2]	*PIN_Y27*	*Seven Segment Digit 5[2]*	*3.3V*
HEX5[3]	*PIN_AB27*	*Seven Segment Digit 5[3]*	*3.3V*
HEX5[4]	*PIN_AB26*	*Seven Segment Digit 5[4]*	*3.3V*
HEX5[5]	*PIN_AA26*	*Seven Segment Digit 5[5]*	*3.3V*
HEX5[6]	*PIN_AA25*	*Seven Segment Digit 5[6]*	*3.3V*

5. Lab Submission Instructions

- The lab is intended to be implemented by each **individual** student. It is also allowed for a **maximum of two students** to work in the lab together. If you decide to work in a group of two students, **both students** must **check off** the working lab assignments with a lab instructor and only **one student** must complete the lab submission as explained in the lab assignment on Canvas.
- Write your lab report following the **report template** provided on Canvas. If you are working with a partner, write both names/IDs on the cover page. It is **prohibited** to include a partner's name on a lab report cover page if that partner does not actively participate in implementing the lab.
- The pre-lab questions prepare you for the challenges you will be presented with in the actual lab. Have their answers ready before the lab session and include these answers in the lab report together with the lab assignments.
- For each lab, submit the following on Canvas before the announced due date/time (only submissions on Canvas are accepted):
    - The lab report as a single Word or PDF document
    - Any files the assignments might ask for (e.g., a demonstration video file)
    - Submit each of the above files separately (do not upload compressed files).
- You can submit multiple attempts for this lab; however, only what you submit in the last attempt will be graded (i.e., all required reports and files must be included in this last attempt).
- **Late submission** of any lab assignment after the given due date and time will be penalized at 1 point per every late hour. No assignment submission will be accepted 48 hours after its due date/time.
- **Successful submission**: it is your responsibility to verify that you submitted successfully all the required files for the lab. First, to verify that your files were submitted, go to the assignment details page, and make sure that it says "Successfully submitted" under grade. Second, you need to validate your submission details by clicking the Submission and Rubric link on that page.
- **Deliberate plagiarism** in any assignment is an extremely serious offense that may result in failing the course. Plagiarism includes submitting answers that are not yours (e.g., copied from someone else or from the Internet). Copying materials from the course slides posted on Canvas is not considered plagiarism.